
Principle and applications of MEMS: a review

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Abstract: Design for miniaturisation represents a broad research topic with applications in fundamental physics, chemistry, martial science, computing methods, ultra-precision engineering, fabrication technology and micromachining based on the principles, characterisation, modelling, simulation and state-of-the-art technology. This paper presents a review on the research and developmental aspects of Microelectromechanical Systems (MEMS). MEMS is relatively a recent development which can be used in a variety of industrial as well as consumer product applications where IC-based technology fails. The review will enable the readers to understand the underlying technology, philosophy, concepts, ideas and principles including micromanufacturing, lithography, bulk and surface micromachining, microsensors, microactuators, Microopto Electromechanical Systems (MOEMS), RF MEMS, Nanoelectro Mechanical Systems (NEMS), microfluidics and BioMEMS. Aspects of microsystems in terms of design process, practice and techniques are presented.

Keywords: MEMS; Fabrication; IC; NEMS; LIGA; Modelling.

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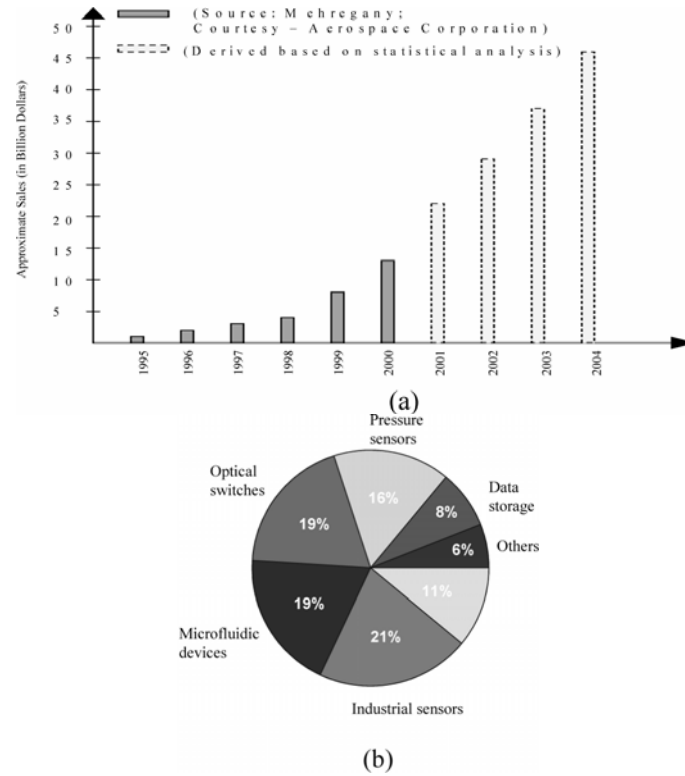
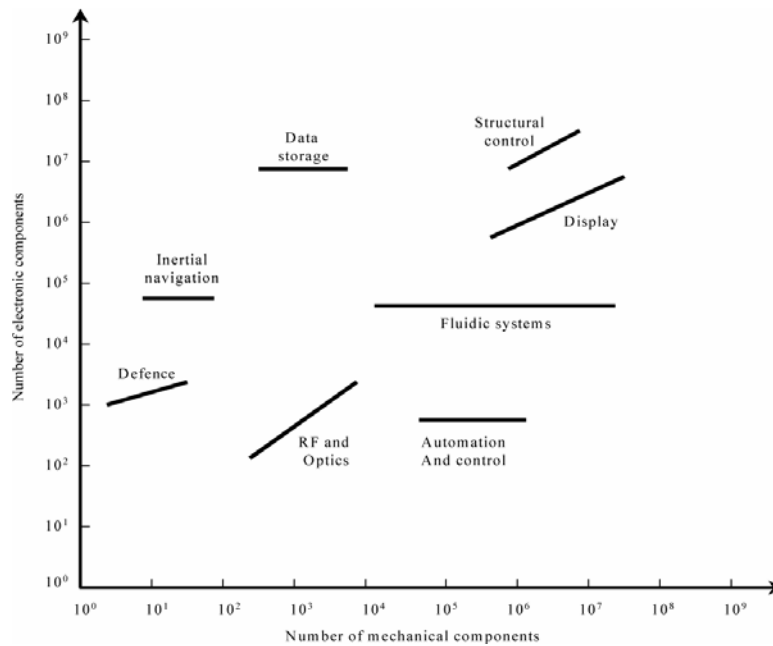
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1 Introduction

Most of the phenomena are to be precisely measured and controlled in a timely predictive manner in order to overcome temporal and spatial limitations. The miniaturised systems have better response time, faster analysis and diagnosis, good statistical results, improved automation possibilities with decreased space, risk and costs. In the year 1959, Feynman proposed the possibility of manufacturing ultraminiaturise systems for a variety of applications, to a level that may involve multiscale formulation methods in terms of manipulation of molecules and atoms (Mahalik, 2007). Sophisticated miniaturised components and systems can change the products and equipments in the most dramatic way (Mahalik, 2005). Microelectromechanical Systems (MEMS) an advanced product and equipment design concept has already emerged in order to cater the need for miniaturisation. Many industries in conjunction with academic institutions and R&D sectors have been investing good amount of money and resources in pursuing technological growth in this field. Currently, MEMS market demand is progressively being strengthened by time-based value engineering that meets the high industry demand (<http://www.aero.org/publications/helvajian/index.html>). The value of MEMS products increased 14 billion USD by the year 2000 that was rightly predicted during 1995. The market for RF MEMS devices, for example, is forecasted to grow to 1 billion USD by 2006. Figure 1(a) shows the illustrations of market growth as far as sales of MEMS products are concerned (www.aspe.net). Figure 1(b) shows approximate percentage of various types of MEMS devices sold in this decade. Figure 2 shows the number of mechanical and electronic components in MEMS devices for typical applications. The design phases of MEMS are highly complex involving multiphysics and inter-interdisciplinary knowledge base.

MEMS devices have already found significant applications in many sectors. They are used for controlling micromanipulator, microequipments, microgrippers and microrobots. Many MEMS devices are found in clock, ink-jet printer head, colour projection and display systems, scanning probe microscopy, to name a few. MEMS technology also found in the design of sensors such as pressure, temperature, vibration, etc. MEMS-based light reflectors, beam splitter, RF and optical switches are common. Broadly the application sectors are:

- 1 aircraft and automotive industries
- 2 chemical industry, manufacturing, clinical and pharmaceuticals
- 3 defence and space applications
- 4 environmental
- 5 communications (wireless, optical)
- 6 pacemakers and data storage devices
- 7 consumer products
- 8 display applications
- 9 inertial navigation devices
- 10 microfluidic applications
- 11 industrial automation and control sectors, etc.

Figure 1 (a) Market growth of MEMS-based products and (b) an approximate illustration with regard to use of MEMS products in various sectors**Figure 2** Density of typical System-on-a-Chip (SOC) devices

2 Mother disciplines of MEMS and NEMS

The R&D activities on miniaturisation of systems broadly fall under two major categories such as MEMS and Nanoelectro Mechanical Systems (NEMS) (<http://www.research.cornell.edu/VPR/vpr.html>, 1998). MEMS technology is considered as amalgamation of two subdomains such as ultra-precision microengineering and IC technology. MEMS and NEMS both are interdisciplinary, multiphysics, multiengineering platforms in which manufacturing of smaller parts, components, products and systems at the level of micro- and nanoscale with more functionalities and capabilities are realised (Denko and Endo, 2002). The former is more close to the mechanical machining processes, while IC technology corresponds to microelectronics design. NEMS groups of systems fall under *nanotechnology* (<http://nanopedia.cwru.edu/PrintPage.php?page=nanotube.chirality>). To some extent, both MEMS and NEMS have been categorised under micromanufacturing, a sister branch of Production Engineering (PE). PE is the cornerstone of many industrial activities that integrates both mechatronics and micromechatronics engineering. Mechatronics is the synergistic integration of mechanical engineering with electronics and intelligent control algorithms in the design and manufacture of products. Technical areas such as motion control, robotics, automotive systems, intelligent control, actuators and sensors, modelling and design, system integration, vibrations and noise control are classified under this field. The synergistic integration of MEMS technology and intelligent control algorithm in the manufacture of products is called micromechatronics.

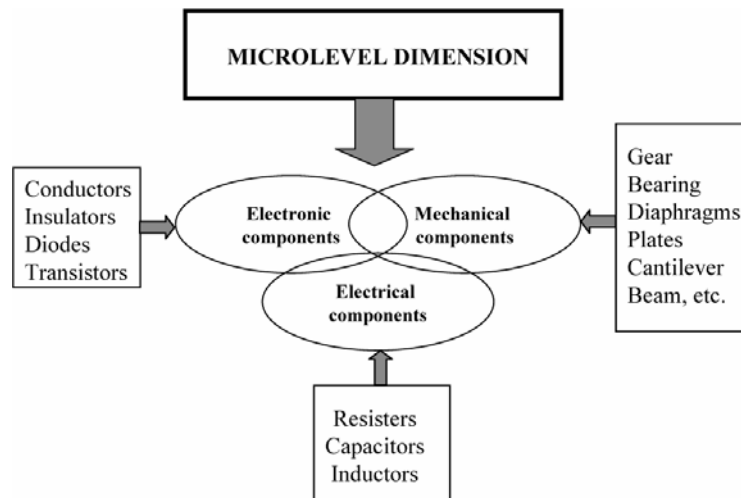
3 Precision, ultra-precision and microengineering

Precision engineering includes design, fabrication and testing where variations in product parameters are caused by logical scientific occurrence. Surface finishing is a precision engineering activity. *Nanomachining* can be considered as an ultra-precision engineering process. It provides advanced material shaping in the nanometer realm. *Micromilling* and *microdrilling* are the two important processes (Jha and Jain, 2005). Traditional design of 3D miniaturised structures is based on micromilling process. Micromilling suggests two different approaches: cutting based and Focused Ion Beam (FIB) based – an advanced method. In the FIB process, for a spot size of 0.45 μm with 2.5 nA of current, the required current density is approximately 1.65 A/cm². Typically, a high milling rate of 0.65 $\mu\text{m}^3/\text{nA s}$, corresponding to an average yield of 6.5 atoms/ion, can be obtained at 45 keV, 30° incidence and 45 scans. Microdrilling refers to drilling of ultrafine holes. 300 nm holes can be successfully drilled on a 1000-Å-thick film using the interfered laser beam compared to micrometer holes using the conventional non-interfered laser beam. The most important parameters of microdrilling process are scaling, accuracy, resolution, sensitivity, repeatability and quality. Aspect ratios up to 100 (e.g. a 15 μm hole in a 1.5-mm-thick foil) in thick material are achievable. High aspect ratio beyond 100 is practically not feasible. The problem can be overcome by utilising Bessel beam (Fischer et al., 2006).

4 IC and MEMS

ICs are Very Large Scale Integrated (VLSI) semiconductor chips. A technology that considers manufacturing of microscale sensors, actuators, valves, gears, mirrors, switches and so on similar to semiconductor chips, are referred to as MEMS (Figure 3). MEMS design technology, an extended form of traditional IC fabrication, can fabricate capacitors and inductors as well as mechanical elements such as springs, gears, beams and diaphragms. It was impossible to fabricate these components utilising IC technology.

Figure 3 MEMS device comprises of electronics, electrical and mechanical elements



4.1 Fabrication processes

The method of micromanufacturing of microelectronic devices is called *fabrication*. The important fabrication sequences are film growth, doping, lithography, etching, dicing and packaging. All these processes are performed on a substrate on which a thin film is grown. Then the properties of the layer are modulated by appropriately introducing *doped* material in a controllable manner. Two commonly used doping methods are *thermal diffusion* and *ion implantation* (http://www.elec.gla.ac.uk/groups/sim_centre/courses/diffusion/diff_5.html). The subsequent process is called lithography, which refers to creation and subsequent transformation of a masking pattern. A *mask*, which is prepared before hand, consists of a glass plate which is coated with a patterned layer usually chromium film. The preparation of mask can be carried out by the use of computer-controlled electron beam to expose the photographic mask material in accordance with the desired configuration and requirement. The pattern on a mask is then transferred to the film by means of a photoresist. The mask features are transferred to the film of the wafer by exposing the light-sensitive photoresist coating through the transparent areas of the mask. The mask then appears on the surface of a thin layer. Once the pattern is transferred into the film, the material areas of the wafer unprotected by the hardened photoresist are then removed by etching. When UV light is used it is called photolithography. X-ray and electron beams are also preferred. *Etching* is a process of removing the portions of a layer of semiconducting material from the base by

chemical or electrolytic means. Etching can be either physical or chemical or a combination. There are broadly two types of etching process: wet and dry etching. In wet etching, the material is dissolved by immersing in a chemical solution (known as etchants), on the other hand in dry etching, the material is dissolved using reactive ions or a vapour phase *etchant*. Some of the commonly used etchants and their etch rate with respect to Si(100), are given in Table 1 (French and Sarro, 1998). Etching techniques are characterised by their selectivity and degree of anisotropy. Then a thin aluminium or gold layer is deposited on the uppermost layers of the chip in order to allow metal to contact the device elements. The aluminium deposition is often achieved by using sputtering method. *Dicing* is a process of cutting up the wafer into individual chips. *Packaging* is a complex process that involves physically locating, connecting and protecting a device.

Table 1 Characteristics of major wet etchants

Etchant	Suitable masks	Etch rate			Comments
		Si(100)(μmmin^{-1}) (100)/(111)	SiO ₂ ($\text{\AA hr}^{-1}$)	SiN ($\text{\AA hr}^{-1}$)	
Hydrazine	SiO ₂ , SiN Most metals	0.5–3 16:1	100	<<100	Highly toxic Explosive at high conc.
EDP	Au, Cr, Ag, Ta, SiO ₂ , SiN	0.3–1.5 35:1	120	60	Highly toxic
KOH	SiN, Au	0.5–2 up to 200:1	1700–3600	<10	Not cleanroom compatible
TMAH+IPA	SiO ₂ , SiN	0.2–1 up to 35:1	<100	<10	Cleanroom compatible

Source: French and Sarroz (1998).

Micromachining refers to the fabrication of 3D MEMS structures with the aid of advanced lithography and etching techniques. Broadly, the processes fall into two categories: bulk and surface micromachining (Bag, 2005). Bulk micromachining refers to etching through both the sides (front and back) to form the desired structures. The structures are formed by wet chemical etching or by Reactive Ion Etching (RIE). The advantage of bulk micromachining is that substrate materials such as quartz or single crystal silicon are readily available and reasonably high aspect-ratio structures can be fabricated. It is also compatible to IC technologies. The disadvantages of bulk micromachining that the process is pattern and structure sensitive and pattern distortion occurs due to different selective etch rates on different crystallographic planes. Further, since both front and backside are used for processing, severe limits and constraints are encountered on the minimum feature size and spacing. Bulk micromachining utilises the etch selectivity between {111} planes and {100} and/or {110} planes in aqueous alkaline etchants. Si wafers with (100) and (110) orientations are used in bulk micromachining. Using Si(100), simple structures such as diaphragms, V-grooves and nozzles and more complex structures such as corner cubes and rectangular masses, can be fabricated. Vertical microstructures can be fabricated on Si(110). It is because Si(110) has four {111} planes that intersect the wafer surface vertically. Surface micromachining is another method that is characterised in terms of fabrication of MEMS structures out of

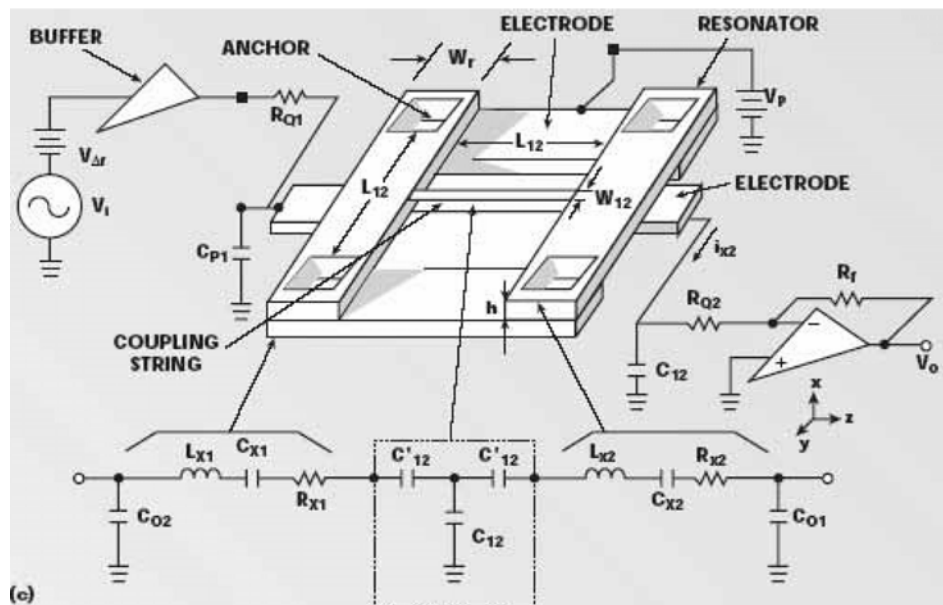
deposited thin films. The thin film may be composed of three layers of materials (Alper and Akin, 2000). The layers are deposited in sequence and subsequently some selective portions of the layer are removed to build up a 3D mechanical structure. Polysilicon, silicon nitride and silicon dioxide are commonly used as the structural, insulation and sacrificial layer, respectively. Hydrofluoric acid can dissolve the sacrificial layer. Once etched, the structure can be freed from the planar substrate. The surface micromachining process is a critical method. Critical issues are:

- 1 basic understanding and control of the material properties of structural films
- 2 releasing method for the microstructure
- 3 fabrication features for hinged structures and high-aspect ratio devices and
- 4 packaging methods.

4.2 SOC concept

MEMS is the integration of active and passive elements on a single silicon substrate. The active elements are sensors and actuators and the passive elements are Passive Electronic Systems (PES) such as signal conditioning circuits (amplifier, ADC, filter, isolators, etc.) and the Passive Mechanical Systems (PMS) such as gear, crank, bearing, etc. IC circuits can be thought of as the nervous system with sensors and actuators as the eyes and arms, respectively. MEMS promise to revolutionise most of the microproducts by combining microfabrication with micromachining processes sequence on silicon making it possible to realise SOC. Figure 4 shows a circuit model of a SOC device, which integrates electronics (amplifier and other passive elements) along with mechanical components such as resonators (De Los Santos and Richards, 2001).

Figure 4 A typical SOC RF resonator



Source: Courtesy – Horizon House Publications, Inc.

4.3 Next generation MEMS

MEMS technology can allow the development of smart systems inheriting additional capability of perception and controlled attributes of microsensors and microactuators. For example, quite large numbers of sensors can be micromachined in a single platform as a sensor fusion device. Sensor fusion is a scheme usually applied in the field of industrial automation and control for their reliable operations. Instead of using a single sensor multiple sensors are employed in order to detect the measurand. A single output is produced based on statistical manipulation. Since many sensors take part in generating a unified output, sensor fusion scheme can eliminate error. The next generation MEMS device can accommodate decision-making capability with embedded soft computing algorithm. Moreover, prognostic measure in terms of sensor and actuator validation (Henry, 1995) can be achieved through next generation advanced design approach.

5 Classifications

Broadly, MEMS are classified based on the principle of sensing, actuation and application domains; accordingly we have Mechanical MEMS, Thermal MEMS, Magnetic MEMS, Microopto Electromechanical Systems (MOEMS), RF MEMS, Microfluidics and BioMEMS.

Micromachined microsensors are designed by adopting various transduction principles such as thermoelectric, photoelectric, electromagnetic, magnetoelectric, thermoelastic, pyroelectric and thermomagnetic (Grimes et al., 2001). The term transducer and sensor are used interchangeably although sensor is considered as a more sophisticated transducer with additional signal conditioning circuits including amplifier, noise eliminator, offset compensator, analogue-to-digital converter or combination of all. In many cases the decision-making logics are also integrated. The decision may be transmitted to the actuators for achieving moving, positioning, regulating, pumping and filtering actions. Sensors vary from very simple to highly complex. They are classified in different ways, such as according to the nature of signal they handle, selective material of the transducer and the level of signal generation. Further, the important attributes such as stimulus, specifications, physical phenomena, conversion mechanism, response, ruggedness, stiffness, range, ability to measure and application field are vital. MEMS sensor used in a typical car are, air-conditioning compression sensor, mass airflow sensor, force sensor, pressure and inertial sensor for braking control, inertial navigation sensor for sensing acceleration, tire pressure sensor, crash sensor, exhaust gas sensor, microphone for noise cancellation, air-bag side impact sensor and fuel sensor (Madou and Morrison, 1989). From this list one can imagine the types of sensors that are expected to be used in other sectors.

Actuator is a device that makes something to move. The process of micromovement is called microactuation. Microactuator are lightweight, conformable and precision devices. The basis of actuation is the method of energy transduction. The microactuation applications are many, including:

- 1 actuation of micromirrors to scan laser beams
- 2 driving of cutting tools for microsurgical applications
- 3 driving of micropumps and valves for analysis

- 4 data reading and recording
- 5 printing applications
- 6 spatial light modulation
- 7 switching
- 8 RF signal tuning, etc. (Koganezawa and Hara, 2001; Lee and Esachi, 1995).

One way to classify the microactuators is based on their movement. It is possible to convert one form of actuation to another, for example, translational to rotational and vice versa, by suitably designing the interfacing components. Anchors, gears, bearings, cams, etc. can be considered interfacing components. Depending upon the load, range and environmental condition the microactuators are selected. Principal energy transduction methods, called driving methods, are mechanical, thermal, electrical and magnetic method. Accordingly the commonly available microactuators are called:

- 1 mechanical actuators (Pandell and Garcia, 1996)
- 2 thermal actuators (Yan et al., 2003)
- 3 electrostatic actuators (Maithripala et al., 2003) and
- 4 magnetic actuators (Shen et al., 2001).

Each driving principle has specific advantages and disadvantages with respect to power requirement, deflection range, required force, environmental durability and most importantly the response time. Of these, mechanical methods currently look promising, although the others have their places (Table 2).

Table 2 Typical power requirements against driving principles

<i>Driving method</i>	<i>Power usage</i>	<i>Voltage range</i>	<i>Current range</i>	<i>Force generated</i>	<i>Deflection range</i>	<i>Speed</i>
Mechanical	High	10–100 V	nA– μ A	Moderate	Moderate	mS
Thermal	High	1–10 V	mA–10mA	Moderate	Small	mS
Electrostatic	Low	10–100 V	nA– μ A	Moderate	Moderate	μ S
Magnetic	Medium	1–5 V	$\sim$ 100 mA	High	Large	μ S–mS

5.1 Mechanical MEMS

Mechanical MEMS mostly emphasises on two classes of devices; mechanical structure based and piezoelectric material-based device. When the geometric structural configurations are exploited for sensing and actuating purpose then the MEMS design can be classified under the first category. Piezoelectric material-based mechanical sensors and actuators exploit the effect of piezoelectricity. Broadly, the mechanical MEMS mechanism utilises.

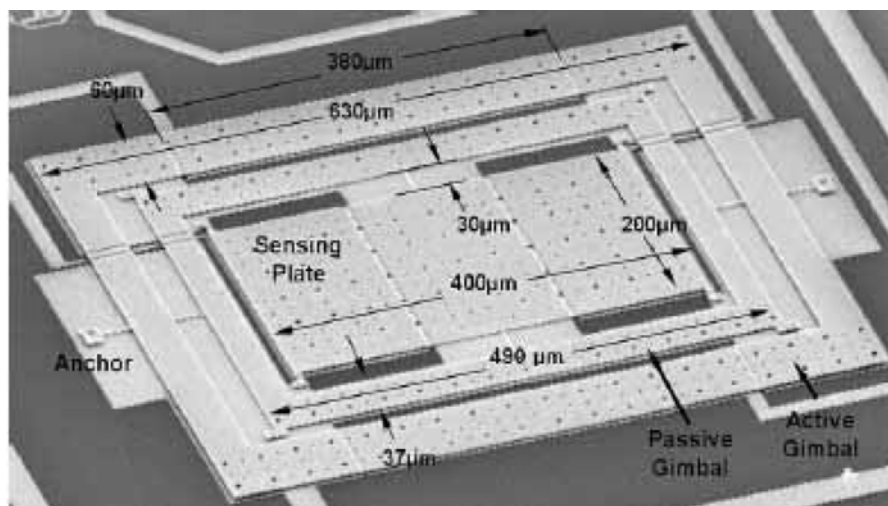
- 1 cantilever beam sensor (Chui et al., 1998)
- 2 capacitive sensing
- 3 microphones (http://microfluids.engin.brown.edu/Breuer_Papers/Conferences/AIAA99-0606_Microphone.pdf)

- 4 gyroscopes (Acar and Shkel, 2004) and
- 5 piezoelectricity-based methods and principles.

Cantilevers bend when pressure is applied and oscillate in a way similar to a spring. The application areas of cantilever sensors are vast including acoustic measurements, vibration monitoring, viscosity and density measurement, infrared and UV radiation detection (Fischer et al., 2006), magnetic and electric field detection, detection of chemical vapours including medical and biological agents, measurement of contaminants in water, explosive vapours, nuclear radiation and detection of DNA (Schienle et al., 2004).

A MEMS microphone (Figure 5) is an ultrasonic acoustic microsensor that transforms acoustical energy into electrical energy and takes the advantages of low power consumption, offering a wide range of applications such as sound detection and analysis, wind noise flow turbulence sensing and vibration sensing and recognition. Microphones are also suitable for outdoors acoustic surveillance on robotic vehicles. Microphones can be arranged in array in order to increase the performance. The principle of operation is based on the vibration of a thin, circular diaphragm held in tension and clamped at the edge and realised in silicon platform. The diaphragm makes up capacitors with the plane that change during the vibration caused by the sound signal. MEMS gyroscopes are designed to measure angular rate of rotation. Typical gyroscope consists of a single mass, oscillating longitudinally with rotation induced lateral deflections being sensed capacitively. The PZT convert mechanical disturbances to electrical signals. Piezoelectric polymers are now being used profusely. Many accelerometers employ piezoelectric sensing technique, utilising cantilever beams.

Figure 5 An SEM of a gyroscope



Source: Acar and Shkel (2004).

5.2 Thermal MEMS

Thermal MEMS work on Peltier effect, thermoresistivity, pyroelectricity and shape memory effect. A voltage/current is developed in a loop containing two dissimilar

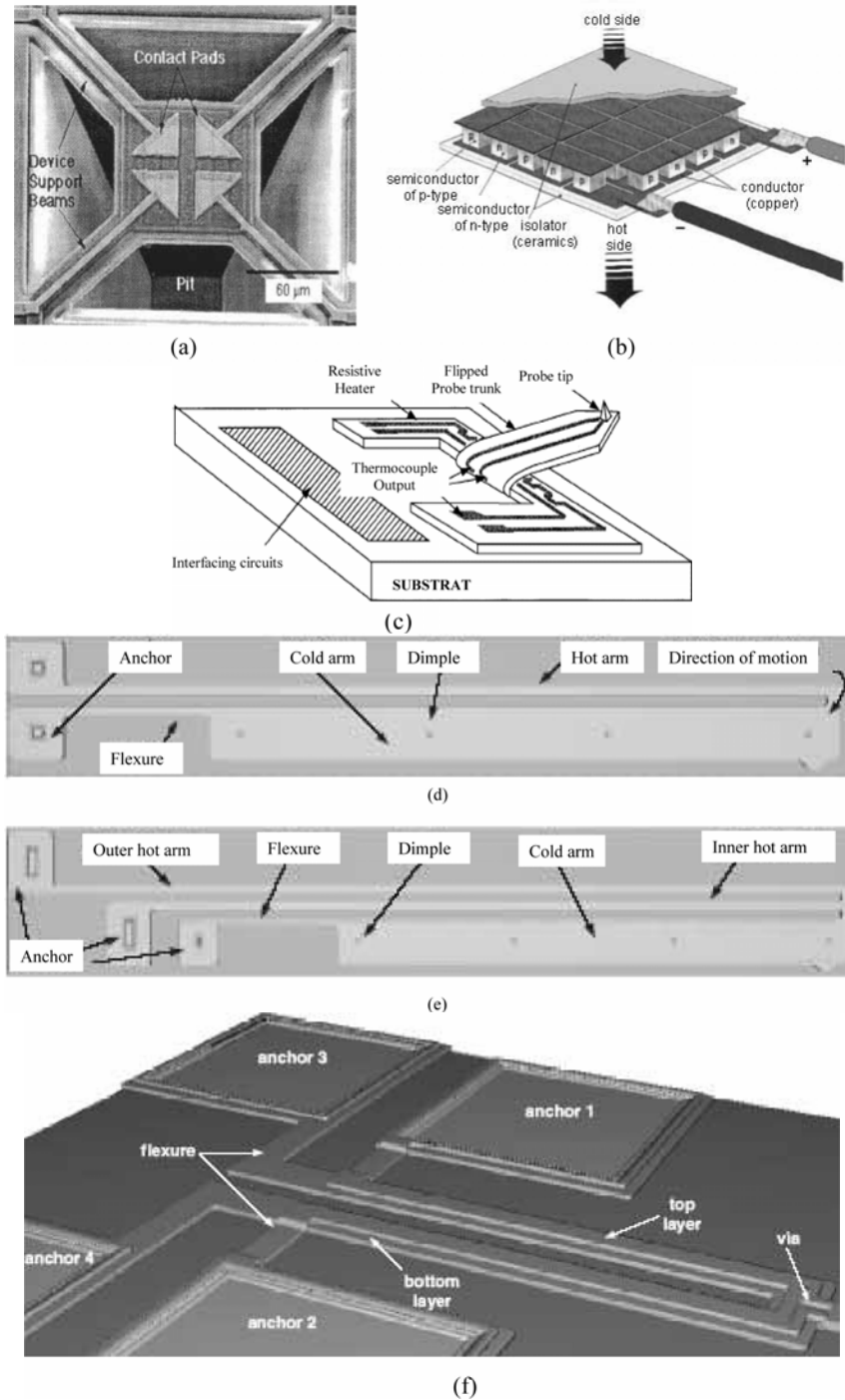
metals, provided the two junctions are maintained at different temperatures. The effect is popularly known as Seebeck effect. The reversed Seebeck effect is Peltier effect. The materials, which produce large range of resistance value to a small range of temperature, are thermoresistive materials. Two types of materials exist: negative temperature coefficient and positive temperature coefficient. Pyroelectricity is the migration of positive and negative charge to opposite ends of the polar axis of a crystal as a result of a change in temperature. Among the materials that reveal the pyroelectric effects are lithium tantalate (LiTaO_3), gallium nitride (GaN), triglycine sulfate (TGS) and polyvinyl fluoride (PVF_2). The shape memory effect is the ability of some specific alloy to recover the original shape under heating. Initially, the alloy is prestrained in low-temperature enabling it to deform severely from its original shape. By applying heat flux the original shape of the alloy can be recovered. The alloy is popularly known as Shape Memory Alloy (SMA). When SMA exhibits the shape memory effects upon both heating and cooling, then it is called Two-way Effect. SMA has two solid-state phases; martensite and austenite phase. Martensite phase is relatively soft and exists at lower temperatures contrast to austenite phase. At this phase the alloy can be easily deformed. In martensite phase the alloy can remain at un-deformed state or at deformed state. Loading can deform the alloy. The temperature at which an SMA undergoes a sphase change from austenite to martensite is called Memory Transformation Temperature (MTT). The most effective and widely used alloys are (Ho et al., 1999) NiTi (Nickel-Titanium), CuZnAl and CuAlNi. The research areas under thermal MEMS includes:

- 1 thermodevices including thermocouple and thermopiles
- 2 peltier heat pump and heat sink devices
- 3 hotwire and microhotplate-based thermal flow sensors
- 4 microthermo-vessels
- 5 U-shaped horizontal and vertical uni- and bidirectional thermal actuators (Figure 6)
- 6 bistable MEMS relays for power system applications and
- 7 thermocouple probe for imaging, topography and data storage applications.

5.3 MOEMS

MOEMS are MEMS but they handle optical signals. MOEMS technology accommodates the principles of optics, electronics and mechanics. MOEMS show good performance with negligible signal degradation and better Quality of Service (QOS) compared to traditional optoelectronic devices. High operational bandwidth and low power consumption are the key features of MOEMS devices. The advantages of MOEMS are (see for more details <http://www-leti.cea.fr/commun/europe/MOEMS/moems.htm>; Sche and Wu, 2004):

Figure 6 (a) A microhot plate as flow sensor, (b) peltier heat pump, (c) a microtip for data recording, (d) U-shaped horizontal and vertical electrothermal actuator, (e) two-hot-arm horizontal thermal actuator and (f) its 3D view



Source: Yan et al. (2003, 2004).

- 1 the speed of operation is high
- 2 high accuracy
- 3 insensitivity to electromagnetic interference
- 4 can be used in harsh environments
- 5 secured, reliability and available
- 6 high sensitivity and selectivity
- 7 mass production capability hence low manufacturing costs.

MOEMS have emerged to provide unparalleled functionality in telecommunication applications. Manufacturers of these devices are forecasting new opportunities in information technology, healthcare, military, industrial and test and measurement sectors. Some of the important applications of MOEMS are (Mahalik, 2007):

- 1 optical sources and photodetectors
- 2 display and projection systems
- 3 optical switches and routers
- 4 tunable sources and filters, reflection modulators, spectral equalisers for optical communications and sensing
- 5 microscanners (image processing, bar code reading, obstacle detection)
- 6 deformable membranes for adaptive optics
- 7 free-space microoptics (diffractive and refractive microlenses) and
- 8 guided optics devices.

LIGA process is primarily adopted for MOEMS devices design (Jian et al., 2003). LIGA is a German acronym for lithography, electroplating and moulding.

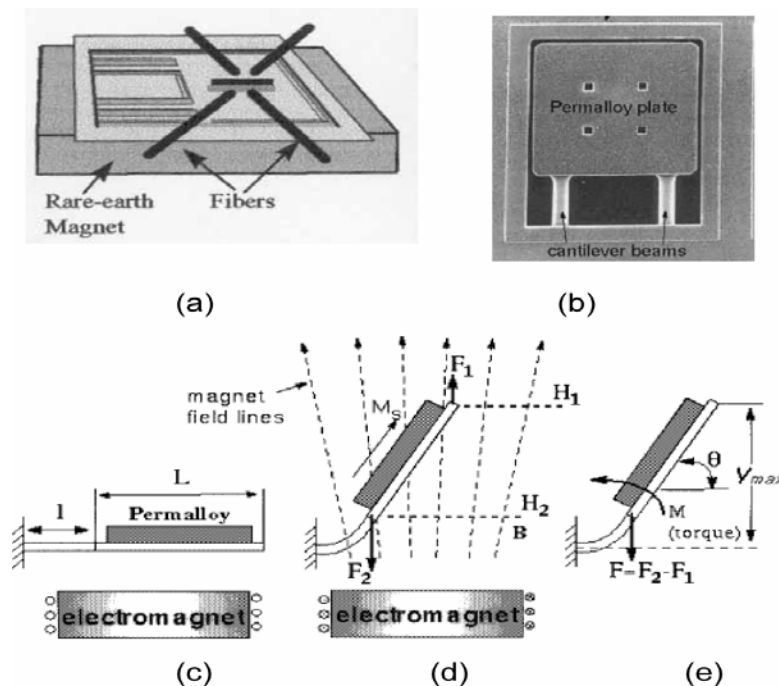
One of the major developments under MOEMS is the Spatial Light Modulator (SLM) (Solgaard, 1992). SLA is a device that modulates light according to a fixed spatial pixel pattern. The general idea of such devices is to modulate optical wave front as a function of input signal. SLM consists of an array of electrostatic parallel plate microactuators that are directly coupled with square tiltable mirror pixels. The important applications are display and imaging.

5.4 *Magnetic MEMS*

Soft ferromagnetic materials have found the most utility in microsensors and microactuators while hard magnetic materials are mostly used for data storage devices. Hard magnetic films with thickness of several microns are grown by sputtering technique. Especially, Magnetoresistive (MR) materials are used for detecting the strength and direction of the magnetic field, which in turn can measure the distance, proximity, position, angle and rotational speed. The MR sensors undergo a change in resistance in response to an applied magnetic field vector. The variation in resistance depends on the rotation of magnetisation relative to the direction of the applied field. The change in resistance of the material is highest if the magnetisation is parallel to the field and lowest if it is perpendicular. Their sensitivity to the direction of the magnetic field

allows for the measurement of angle also. For the development of micromachined magnetic sensor nickel-iron permalloy are widely used. Permalloy has a low magnetic anisotropy. The anisotropy field is about 3 and 5 Oersted. Actuators are capable of generating large bidirectional forces with long working lengths. Deflections of over 1000 micron can be achieved with surface micromachined cantilevers, magnets and pancake coils. Magnets with typical volumes measuring in the range of 0.11 mm^3 can produce forces in the mN range. A typical magnetically actuated microcantilever incorporating a magnet is measured $2000 \times 1000 \times 100$ microns (Figure 7). Deflection at the tip of the cantilever can be about 1000 microns subjected to a driving current of 100 mA passed through a 40 turn pancake coil (Guo-Ming, 2004). The actuation force depends on magnetic field intensity, the magnetic susceptibility and the mass of magnetic susceptible material. Large deflections causes non-linear performance however, small deflections show better linear current-deflection relationship. The actuator can be uni- or bidirectional type depending upon the way the actuation takes place (Cho and Ahn, 2002; Wright et al., 1997). As name suggests the bidirectional actuators can drive in positive and negative direction with respect to reference.

Figure 7 (a) Magnetic 2×2 optical bypass switch, (b) two cantilever-based paddle type magnetic actuator and (c-e) schematic of magnetic actuation



5.5 RF MEMS

RF MEMS add new capabilities and improved power efficiency. Devices used for RF communication are switches, inductors, varactors, filters, tuners and resonators: the important components of a typical mobile phone. MEMS versions of these components promise to make devices more reliable and power efficient. RF MEMS can be used for achieving:

- 1 transmission and reception
- 2 VCO tuning
- 3 RF band select filters and Intermediate Frequency (IF) filtering
- 4 time delay for phased-arrays
- 5 Variable Delay Lines (VDL) and
- 6 reconfigurable antennas.

5.6 *Microfluidic systems*

Microdevices, which are used to transport and store fluid, are called Microfluidic Systems (MFS). Typically the MFS handle fluid volumes on the order of nanoliter. Some of the important building blocks of MFS are:

- 1 microchannel
- 2 micronozzles
- 3 micropumps
- 4 microvalves and
- 5 microreservoirs (microvessels).

Some important applications are inkjet printing, drug dispensing, reaction analysis, mixing and separation, chemical synthesis, detection of chemical species, genetic analysis and semiconductor processing. The advantages of MEMS compared to conventional fluidic systems are that the miniaturised system requires less reagent (species or samples) resulting in faster, accurate and reliable measurements. MFS significantly differs from macroscale hardware as the behaviour of fluid at microlevel is considerable. For instance, the capillary action changes the way in which fluids pass through microscale diameter channels, as compared with macroscale tubes. As the dimensions of a device reach a certain size as the scale becomes smaller, the fluid particles or the solvent become comparable in size with the channel or the device itself. MEMS-based technology can create pressure regulators, shut-off valves and Mass Flow Controllers (MFC) for Electronics Speciality Gases (ESG) distribution. The use of MFS and modules eliminates the size and number of welds and face seals used in integrated gas control and distribution components. Regardless of the application domains of MFS, the technological design issues to be addressed are:

- 1 precision alignment, accuracy, geometrical regularity and smoothness
- 2 mechanical parameters such as resistance to moderate and high pressures
- 3 architecture for complex structure and packaging density and
- 4 standardisation issues.

5.7 *Bio and chemo devices*

Microdevices used for analysis and detection of biomedical and industrial reagents are called bio- and chemodevices. Unlike MFS bio- and chemodevices are diode-type, transistor-type or 3D cantilever structure. Some of the applications are:

- 1 forensics and genetic screening
- 2 stress-response analysis
- 3 antibodies gene expression in transgenic cells
- 4 identification of patients with high tumour risk
- 5 biowarfare agents
- 6 pathogens like throat bacteria and
- 7 drug discovery.

The development of DNA sensors is considered as the most innovative molecular biology technology (Liu and Lu, 2003). These micro- or nanosystems can allow an easy, accurate, fast and reliable way to analyse many DNAs simultaneously. The basic element of the DNA analysis system is the DNA probe(s). The technology is based on the recognition of oligonucleotides by their complementary genomic sequences. The process of recognition is called *hybridisation*. The hybridisation can be achieved in three ways such as optical method, surface stress and electrical detection. Optical method is a traditional method, in which the colour of the solution, in some cases change. Surface stress-based hybridisation process uses a probe of microcantilever structure, coated with the detector film that reacts with the biomolecules of interest. A biochemical reaction at the cantilever surface changes in the surface stress. The surface stress causes a bending of the cantilever. A piezoresistor integrated with the cantilever sense the bending variable. The electrical characteristic-based process uses microelectrodes. There are two opposite electrodes and the electrical resistance between the electrodes is infinite. Upon hybridisation, a bridge like structure is formed as a result the electrical resistance between the electrodes drops. The change in resistance value is the measure of presence of DNA. One can note that the sensor is a microstructure, but the hybridisation phenomenon is at the nanoscale level.

6 NEMS

The nanoscale devices and equipments can give added benefits in terms of further miniaturisation and efficiency. Nanotechnology has accelerated research and development in many disciplines. It finds abundance applications in many fields including automotive, aerospace, household appliances, sporting goods, telecommunications equipment and medical. A key obstacle to the development remains the need for cost-effective large-scale production methods (Chen and Huang, 2005). Fullerenes and nanotubes are found to be useful for the nanoscale components and systems. Much of current researches focus on potential applications of Carbon Nanotubes (CNT). CNT are considered as the ultra-fine unique devices, which can offer significant advantages over many existing materials due to their remarkable mechanical, electronic and chemical properties. With strong covalent bonding they possess unique one-dimensional structure. Nanotubes can be utilised as electronics devices, super-capacitors, lithium ion batteries, field emission displays, fuel cells, actuators, chemical and biological sensors. CNT typically are longer in length, usually measuring from about a few tens of nanometers to several micrometers with diameter can be up to 30 nm and can be as small as 3 nm. Each nanotube is a single molecule made up of a hexagonal

network of covalently bonded carbon atoms. Freestanding CNT can be grown by Chemical Vapour Deposition (CVD). They possess metallic to semi-conducting properties along with good thermal conductivity. Other essential properties they possess are high:

- 1 tensile strength
- 2 resilience and
- 3 current densities.

Various structural constructions for a variety of applications can be formed through appropriate methods. Sputtering of carbon atoms from the nanotubes lead to dimensional changes. The dimensional change facilitates surface reconstruction with annealing. X-like junction with diverse angles between the branches can be formed. Under careful irradiation, one of the branches of the X-junction can be removed, creating Y- and T-like junction. These tubular junctions exhibit intrinsic non-linear transport behaviour, depending mainly on geometrical configuration and on the kind of the topology (Tans et al., 1997). Different junctions show different current versus voltage characteristics giving rise to the possibility of using as nanoscale electronic devices for *nanocomputers* (Keren et al., 2003). Nanotube-based design scenario even expects the design of gears and bearings and hence the machines at the molecular level. A fuel cell (Iyuke et al., 2003) is an electrochemical system that generates electricity by converting chemical energy into electrical energy. The fuel cell can be thought of as high-efficient next generation micro power system for variety of applications. Several challenges to widespread implementation of fuel cell technology are needed, although novel inexpensive and long-lasting electrocatalyst materials are becoming major factors in design and development. Recent advances in the application of nanostructured carbon-based materials have suggested the possibility of using CNT as novel electrocatalyst supports for fuel cell.

7 Modelling and simulation

Prior to their design, it is desirable to study the behaviour of the MEMS system. Simulation is performed by the use of computer-assisted tools. Computer-assisted simulation tools contain all types of mathematical building blocks so that the designer can formulate any kind of model equations looking at the system and can subsequently analyse and predict the behaviour. Even the designer can incorporate the physical properties (bulk modulus, permittivity, coefficient of resistance, etc.) of the material into the model equations. Available popular simulation tools for MEMS modelling and analysis are ansys, sugar, etc. (Cole et al., 2003; Lin et al., 2004).

8 MEMS packaging and design considerations

Much like IC packaging, MEMS packages must have the ability to meet some important criteria (Lee et al., 1998), such as:

- 1 there should be good isolation between the non-sensing and sensing areas of the device

- 2 there must not be any hindrance to the driving actions such as tilting, twisting, rotating, sliding or vibrating
- 3 efficient coupling at the link, junction, anchor area
- 4 unreliability issues due to contamination, fusing, sticking, clamping, static overload, de-lamination, creep and fatigue.

9 Conclusion

This paper reviewed the principle and application aspects of MEMS. The scope of MEMS includes:

- 1 microfabrication and micromachining (bulk and surface and micromachining)
- 2 lumped-parameter-based system description and modelling
- 3 mechanical, electrical and thermal properties of material
- 4 SOC and related mechanical and electronic passive systems
- 5 mechanical MEMS
- 6 thermal sensors and actuators
- 7 MOEMS
- 8 magnetic sensors and actuators
- 9 radio frequency MEMS
- 10 MFS
- 11 biomedical and chemical microsystems
- 12 nanotechnology
- 13 simulation and multiscale design approach
- 14 performance measuring parameters and
- 15 devise arrays.

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